

Towards autonomous scheduling agents for water distribution networks: Self-evolving data-centric AI via continuous data-model coevolution

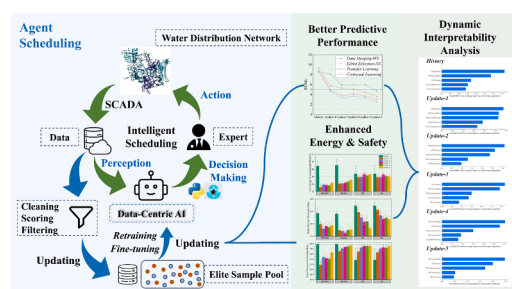
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HIGHLIGHTS

- A self-evolving DCAI agent for intelligent optimal scheduling of large-scale WDNs.
- Continuous model updates with dynamic data curation achieve over 70% MAE reduction.
- 4% energy and pressure fluctuation reduction with over 99% pressure satisfaction.
- Evolution of SHAP values validates directional controllability of optimization.
- Active learning enhances fine-tuning with curated data, enabling flexible deployment.

GRAPHICAL ABSTRACT



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ABSTRACT

Urban water distribution networks (WDNs) require adaptive scheduling strategies to handle dynamic demand fluctuations. Scientific pump scheduling ensures reliable water supply while minimizing energy consumption and carbon emissions. Data-centric AI (DCAI) enables real-time intelligent scheduling of large-scale WDNs by learning human expertise from historical data, yet its effectiveness is often constrained by data quality and static model deployment. Most existing studies train models once for long-term use, neglecting continuous online updates and limiting practical adaptability; this constitutes a critical barrier for autonomous scheduling agents. To address this issue, this study proposes a data-model coevolution mechanism that transforms DCAI into an autonomous scheduling agent with two core capabilities: (1) a dynamic data curation system that ensures high-quality data streams through multi-criteria filtering, and (2) a data-coevolving model update mechanism with retraining or fine-tuning options for continuous online learning and self-evolution. Validated on a real network in Shanghai over 10-week cycles, the agent-based mechanism reduced MAE by up to 71.03% and improved water supply security and energy efficiency. Interpretability analysis confirmed its directional controllability during autonomous updates. Furthermore, when extended with an active learning variant, the proposed approach achieved greater than 25% MAE reduction across scenarios, addressing the inherent challenge faced by transfer and continual learning approaches in capturing minority operational patterns. By endowing DCAI with autonomous online learning capabilities, this work establishes a synergistic paradigm of intelligent scheduling in which the agent achieves automated adaptive updates through interaction with the WDN, thereby generating superior scheduling strategies.

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1. Introduction

Pump scheduling constitutes a cornerstone for ensuring hydraulic reliability and operational efficiency in water distribution networks (WDNs). Time-varying demand profiles induce significant pressure fluctuations, where insufficient pressure compromises service delivery while excessive pressure elevates risks of pipe leakage (Li et al., 2024) and bursts (Xu et al., 2020), and systemic energy losses. Consequently, precision pressure control through optimized pump scheduling is imperative. Furthermore, pumping operations account for up to 70% of total WDN energy consumption (Smith et al., 2019; Janus et al., 2023), with associated greenhouse gas emissions representing a primary carbon source in water systems (Gui et al., 2024; Sugimura et al., 2024). Thus, pump scheduling represents a pivotal opportunity for energy conservation and emission reduction, where strategic operational planning enhances hydraulic efficiency while minimizing energy intensity.

Operator-driven scheduling methodologies leverage empirical expertise to achieve implementation simplicity and rapid emergency response. However, this experience-dependent paradigm suffers from significant subjective bias. Consequently, extensive research has pursued optimal scheduling frameworks to minimize energy consumption while ensuring hydraulic reliability, a domain predominantly governed by model predictive control (MPC) methodologies (Darby et al., 2012; Liu et al., 2019). The MPC workflow comprises three sequential phases: (1) demand forecasting: utilizing predictive models (Jowitt et al., 1992; Antunes et al., 2018; Hu et al., 2019; Smolak et al., 2020; Chen et al., 2022); (2) hydraulic simulation: implementing either macroscopic (Demoyer et al., 1975; Zhou et al., 2022; Fu et al., 2022) or microscopic models (Walski et al., 2001; Ateş, 2017; Hossain et al., 2021); (3) optimization execution: employing mathematical programming (Ormsbee et al., 1994; Giacomello et al., 2013; Skworcow et al., 2014) or heuristic algorithms (Babaei et al., 2015; Makaremi et al., 2017; Rajabpour et al., 2020). While MPC generates more cost-effective and operationally safer schedules than conventional approaches, its computational complexity impedes feasible real-time deployment, particularly in large-scale networks.

The goal of real-time intelligent scheduling is to transition from experience-driven manual operations to autonomous, self-evolving AI agents capable of real-time perception, decision-making, and continuous adaptation. This shift aligns with broader AI trends that integrate large language models (LLMs) (Wang et al., 2025) and foundation models (Wang et al., 2026) into agent-based frameworks for complex system control, enabling reasoning, planning, and human-like interaction. These requirements impose high demands on the response speed of the core algorithms. Reinforcement learning (Hajgató et al., 2020; Xu et al., 2021; Hu et al., 2022) and deep learning (Pu et al., 2023) are two efficient approaches for building such agents due to their computational efficiency. The former tends to explore actively for superior strategies, while the latter focuses on learning knowledge embedded in historical data. However, reinforcement learning's convergence and decision performance heavily depend on accurate hydraulic models, limiting its applicability in large-scale real-world networks. Meanwhile, most urban WDNs are equipped with Supervisory Control and Data Acquisition (SCADA) systems that collect extensive monitoring data, including historical scheduling records, network states, and outcomes. Deep learning leverages data mining capabilities (LeCun et al., 2015) to extract actionable insights from SCADA data and operational expertise without hydraulic models, making it well-suited for large-scale networks. Its high transferability further facilitates method generalization and deployment. Nevertheless, model performance remains critically dependent on data quality (Chen et al., 2014). As research on model architecture reaches a bottleneck, attention has shifted toward improving data quality (Baensens et al., 2021), establishing data-centric AI (DCAI) as the dominant paradigm (Jakubik et al., 2024). This trend also holds in water engineering applications (Fu et al., 2024). Training sets contaminated by low-quality patterns impair knowledge

acquisition, whereas incorporating feasible strategies enhances information extraction.

Constructing DCAI-based core of intelligent scheduling agents requires addressing two fundamental challenges: developing historical data-informed DCAI models to replace manual scheduling, and continuously refining these models for superior decisions. However, most current research remains confined to static models trained once for long-term deployment, lacking self-evolution capabilities, which is a key ability for agents to continuously adapt to changing environments (Gao et al., 2025). While increasing high-quality data enhances initial model performance, poor overall data quality renders model optimization insufficient and severely limits practical AI deployment. Furthermore, static models struggle to adapt to continuously changing operating conditions, necessitating the empowerment of continuous learning and evolution capabilities to transform static predictors into autonomous agents for intelligent scheduling.

To enable DCAI-driven agent scheduling, this study implements a three-phase methodology covering initial model construction through autonomous online optimization:

- (1). Development of a hybrid Conv1D-GRU neural architecture for scheduling action prediction as the agent's core decision module;
- (2). Design of a dynamic data curation system incorporating multi-criteria scoring, quality filtering, and incremental update mechanisms to enable agent memory and experience refinement;
- (3). Refining models via retraining or fine-tuning within a data-model coevolution mechanism ensuring rapid adaptation, a capability essential for agents in dynamic environments.

2. Methodology

2.1. Autonomous scheduling agents for WDNs

2.1.1. Agent scheduling workflow

An agent must possess the fundamental capabilities to operate autonomously, perceive and respond to environmental changes, take proactive actions to achieve goals, and interact with humans or other agents (Wooldridge and Jennings, 1995). In the specific context of intelligent scheduling for WDNs, the agent performs a continuous cycle of perception, decision making, and action under both human-machine interaction and automated operation, as illustrated in Fig. 1:

- (1). Perception: The agent acquires real-time network monitoring data via the connected SCADA system, and processes and stores the data through the data curation system.
- (2). Decision making: The AI model, having learned scheduling knowledge, analyzes real-time WDN data to predict scheduling actions and generate control strategies. Hydraulic simulators are used to simulate the impact of the proposed strategies on the network, enabling evaluation and selection of the most suitable strategy, which is then pushed to the front-end operation platform. This study implements online hydraulic simulation using the WNTR Python library (Klise et al., 2017), which is based on the EPANET 2.2 kernel (Rossman, 2020).
- (3). Action: Front-end technicians verify the reasonableness of the selected strategy, execute the physical control actions, and provide feedback to the agent.

It is worth noting that, each scheduling decision directly affects regional water supply security, every control action generated by the intelligent scheduling agent, particularly during the initial deployment phase, must pass through a rigorous multi-stage verification pipeline, which includes online hydraulic simulation, comprehensive strategy evaluation, and manual review. The outcomes of this manual review are promptly fed back to the backend, thereby establishing a closed-loop data cycle.

This study addresses the real-time scheduling of treated-water pumps that convey purified water from the plant into the WDN. The core scheduling task is to predict the next-period pump operational parameters (frequency or on/off status) from the current hydraulic state, to maintain the required service pressure while minimizing energy consumption. At the target plant, the pumps are governed by a closed-loop PID control system that modulates their speed according to a prescribed pressure setpoint; therefore, the control pressure serves as the primary decision variable optimized by the scheduling agent (Mays, 2000). For adaptive pressure management, the agent issues updated control-pressure commands every five minutes. The agent's output interface is designed to be flexible and can be readily reconfigured for alternative control variables, such as direct pump frequency modulation.

2.1.2. Self-evolution of scheduling agents

A static agent cannot adapt to ever-changing environments; therefore, a truly practical agent must possess the capability for self-evolution through interaction. As illustrated in Fig. 2, after establishing the initial DCAI model (Section 2.2), we integrate the data management method (Section 2.3) and model updating method to construct a systematic approach for data-model co-evolution (Section 2.4). During the agent-WDN interaction process for intelligent scheduling, the agent can autonomously complete adaptive updates.

2.2. DCAI model

A deep learning neural network is established to map inputs (historical monitoring data) to outputs (predicted control pressure). The network structure consists of one-dimensional convolution (Conv1D) layers and gated recurrent unit (GRU) layers in parallel. This model is able to process data at different time scales and extract spatiotemporal information simultaneously. This capability has been validated through real-world engineering applications (Pu et al., 2023). Details of the specific neural network architecture, input/output settings, and hyperparameter selection (Bergstra et al., 2012) are provided in the Supplementary Material.

2.2.1. Evaluation indicators

Model performance is assessed from three perspectives:

(1). Prediction accuracy of the deep neural network, quantified by mean absolute error (MAE) and root-mean-square error (RMSE).

$$MAE = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i| \quad (1)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2} \quad (2)$$

where Y_i is the observed value and $\hat{Y}_i$ is the forecast value.

(2). Energy consumption of the pumps after the proposed scheduling strategies are implemented.

$$E = \frac{\sum_{i=1}^n \rho g H_i Q_i t_i / \eta_i}{3600 \sum_{i=1}^n Q_i t_i} \quad (\text{kWh} / \text{km}^3) \quad (3)$$

where ρ is the density of water (1000 kg/m³); g is the gravitational acceleration (9.8 m/s²); i is the index of the i th time interval among n total intervals; t_i is the duration of interval i (h); H_i is the control pressure at the water plant during interval i (mH₂O); Q_i is the flow rate at the water plant during interval i (m³/h); η_i is the pump efficiency during interval i .

(3). Pressure at the critical monitoring node after the scheduling strategies are applied, evaluated by the hydraulic model. The pressure satisfaction rate r_s and pressure fluctuation rate r_f are defined as:

$$r_s = \frac{n_s}{n} \times 100\% \quad (4)$$

where n_s is the number of time steps in which the nodal pressure meets the minimum requirement; n is the total number of sampling instants.

$$r_f = \frac{\sum_{t=1}^n \delta(P^t - P^{t-1})}{n} \quad (5)$$

$$\delta(P^t - P^{t-1}) = \begin{cases} 0, & |P^t - P^{t-1}| \leq \Delta P_p \\ 1, & |P^t - P^{t-1}| > \Delta P_p \end{cases} \quad (6)$$

where P^t is the nodal pressure at time t (kPa); ΔP_p is the allowable pressure fluctuation amplitude, which is set to 5 kPa in this study based on water supply safety requirements.

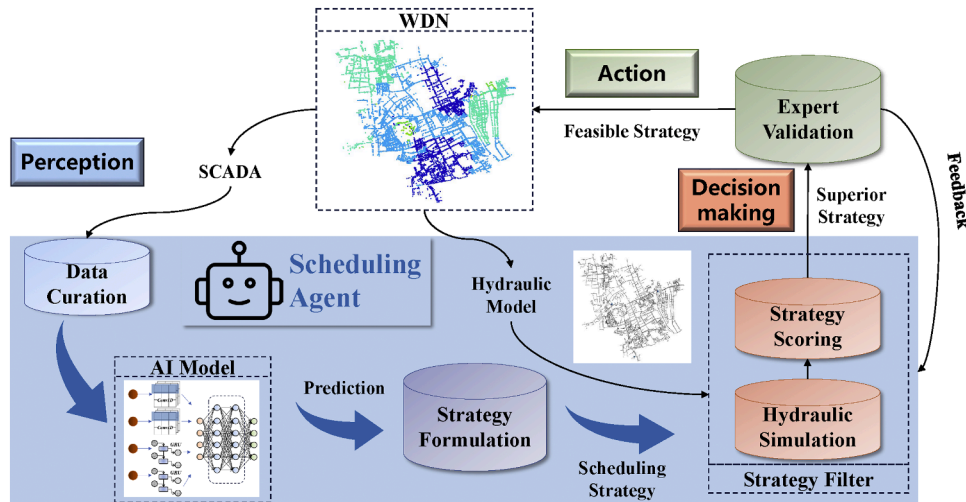


Fig. 1. Workflow of agent scheduling.

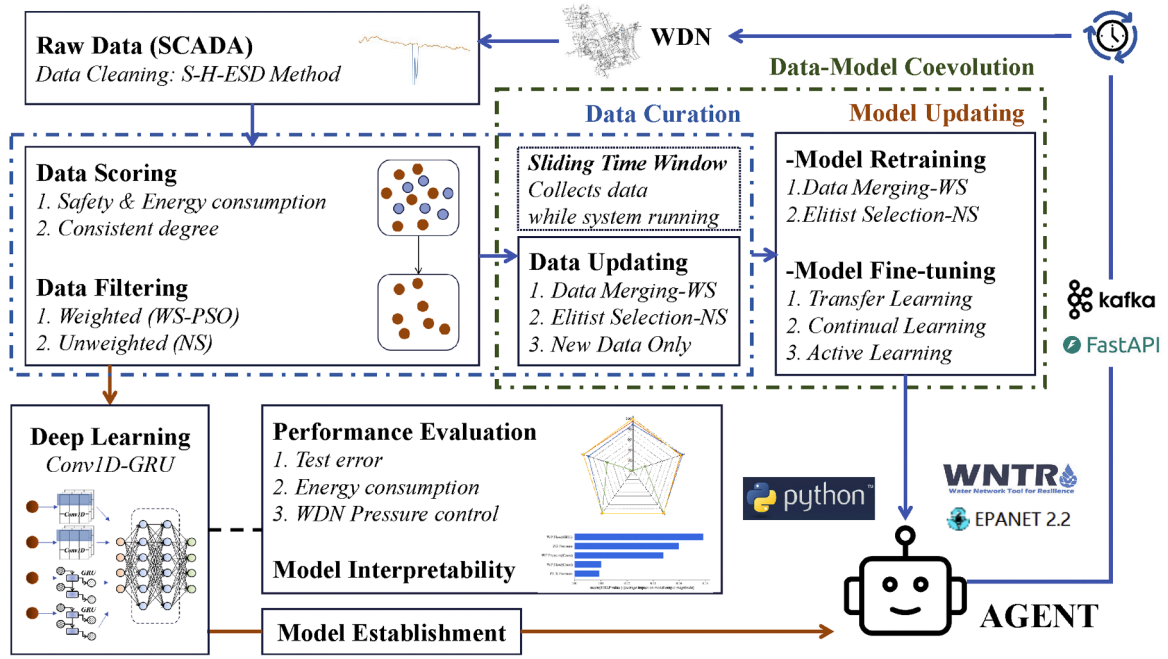


Fig. 2. Self-evolution process of the scheduling agent.

2.2.2. Model interpretability

During practical deployment, the black-box nature of neural networks often undermines trust in their reliability, making interpretability analysis essential for clarifying how input features influence model outputs. Common methods include SHAP (SHapley Additive Explanations) (Lundberg and Lee, 2017) and LIME (Local Interpretable Model-Agnostic Explanations) (Ribeiro et al., 2016). In this study, we adopt SHAP to analyze whether the contribution of each input feature to the output aligns with fundamental principles of WDN scheduling.

The explanation model $g(x')$ belongs to the class of additive feature attribution methods. The prediction for a single instance can be precisely decomposed into a baseline value ϕ_0 (the average prediction when all inputs are unknown) plus the sum of individual feature contributions:

$$g(x') = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (7)$$

$$x = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (8)$$

where M is the total number of features, x' is the simplified input with $x' \subseteq \{0, 1\}^M$, and ϕ_i (the Shapley value) represents the contribution of feature i . The fundamental formula for computing ϕ_i is:

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)] \quad (9)$$

Here, $z' \subseteq \{0, 1\}^M$, $|z'|$ denotes the number of non-zero elements in z' , and $z' \setminus i$ represents setting $z'_i = 0$. A higher contribution (importance) of a variable corresponds to a larger Shapley value.

2.3. Data curation for DCAI

The dynamic data curation system primarily encompasses three basic mechanisms: data scoring, data filtering and data updating. Data updating is closely linked to the data-model coevolution mechanism and will be detailed in Section 2.4.

2.3.1. Data scoring

Historical data were structured into discrete scheduling episodes. We cleaned the raw data using the seasonal hybrid extreme studentized deviate (S-H-ESD) method (Rosner, 1983).

A dual Gaussian kernel-based (Schölkopf et al., 1997) metric (s_1, s_2) evaluates each episode's operational compliance and strategic consistency. The s_1 metric quantifies strategy rationality and efficacy through pre-operational nodal pressure at critical points and post-operational energy expenditure. The s_2 indicator specifically assesses the congruence between proposed scheduling strategies and historically validated manual operations under comparable operating conditions. The formulas for s_1 and s_2 are provided in the Supplementary Material.

2.3.2. Data filtering

Two types of data filtering are employed, depending on whether weights are assigned to the evaluation metrics: weighted sorting (WS) and unweighted sorting (US).

(1). The weighted sorting (WS) approach assigns distinct weights to the evaluation metrics and computes a composite score as follows:

$$S = w_1 s_1 + w_2 s_2 \quad (10)$$

where w_1 and w_2 are the weights assigned to s_1 and s_2 , with $w_1 + w_2 = 1$. A historical scheduling strategy is retained if its composite score S exceeds the threshold S_t .

Unlike the subjective weighting method (SWM), which relies on manual tuning, and the entropy weight method (EWM) (Zou et al., 2006), which isolates weight determination, we adopt particle swarm optimization (PSO) (Marini et al., 2015) for weighted sorting (WS-PSO). This enables efficient, automated determination of both the weights w_i and the threshold S_t . In the particle swarm optimization algorithm, the fitness function f is set as:

$$f = \exp\left(-\frac{\|H_l - \text{mean}(H)\|}{k_1}\right) + \left(1 - \frac{\text{var}(P_{wp})}{k_2}\right) + \left(1 - \frac{\text{var}(P_{point})}{k_3}\right) + \left(1 - \frac{E}{k_4}\right) \quad (11)$$

where $\text{mean}(H)$ is the average pressure at the critical monitoring point (kPa); $\text{var}(P_{wp})$ and $\text{var}(P_{point})$ denote the variances of the control pressure and the critical-point pressure, respectively; E represents the energy consumption per unit of water supply (kWh/km^3); k_1, k_2, k_3 and k_4 are scaling parameters experimentally set to 25, 100, 100, and 100.

(2). The unweighted sorting (UWS) approach treats all metrics equally. We adopt non-dominated sorting (Deb et al., 2002), which hierarchically categorizes solutions based on dominance to identify the Pareto-optimal frontier. The highest-ranked non-dominated tier forms the comprehensive Pareto-optimal set, exhibiting superior multi-objective performance.

For solutions within the same tier, crowding distance (d_i) distinguishes them:

$$d_i = \sum_s \frac{f_{i+1,s} - f_{i-1,s}}{f_{\max,s} - f_{\min,s}} \quad (12)$$

where i indexes solutions sorted from smallest to largest; s denotes objective dimensions; $f_{i,s}$ is the value of the i th solution in dimension s .

Non-dominated sorting proceeds in two stages: tier-based stratification followed by diversity optimization. First, n_1 candidates are extracted from the top k Pareto tiers. Then, n_2 solutions are selected from the $(k+1)$ th tier by maximizing crowding distance d to preserve diversity. This mechanism, denoted NS, serves as the unweighted multi-criteria filtering method in this study.

2.4. Mechanism of data-model coevolution

As shown in Fig. 3 (dashed lines indicate optional paths), the model and data coevolve through the continuous use of the human-in-the-loop intelligent agent scheduling. After the initial model is embedded into the scheduling agent, the generated decision schemes are applied to the WDN following dual validation: hydraulic simulation and performance evaluation, together with expert verification. Feedback from the WDN then yields new and improved scheduling samples that better adapt to current operating conditions. These new samples are subsequently scored and filtered by the data curation system to update the training sample pool, thereby increasing the proportion of high-quality data. Using the improved sample pool, the model can be updated in two

optional ways: retraining with both historical and new data, or fine-tuning that uses only new data while retaining historical information to preserve memory. The optimized model produces superior scheduling schemes and is re-embedded into the scheduling agent for the next cycle. Through this iterative process, the coevolution of the scheduling database and the model is achieved.

2.4.1. Model retraining with historical data retention

This coevolution process systematically integrates historical and newly validated data into optimized training sets through two evolutionary strategies:

Method (1) DM-WS (data merging with weighted sorting). During iterative updates, the sliding time window triggers full-spectrum re-evaluation of all candidate data via the WS-PSO algorithm (Section 2.3.2 (1)). This method enforces rigorous quality control by dynamically reweighting s_1 and s_2 metrics across the unified dataset, as visualized in Fig. 4(a).

Method (2) ES-NS (elitist selection via non-dominated sorting). Implementing the elitist selection strategy (Deb et al., 2002), this approach selectively retains high-performance historical data while assimilating novel instances. The combined pool undergoes NS filtering (Section 2.3.2 (2)) to extract n Pareto-optimal solutions per update cycle (Fig. 4(b)), ensuring diversity-preserving excellence inheritance.

2.4.2. Model fine-tuning with new-scenario data

We introduce transfer learning (TL) and continual learning (CL) to achieve model fine-tuning in this research. The basic idea of TL (Pan et al., 2010) and CL (Parisi et al., 2019) is to utilize new data to fine-tune the parameters of the neural network with a smaller learning rate and fewer training epochs, so as to enable the adaptation of the model to novel application scenarios (Tajbakhsh et al., 2016). This strategy bypasses computationally intensive retraining while obviating historical data retention requirements.

A critical distinction between TL and CL lies in knowledge preservation: TL prioritizes target task performance without explicit memory retention, whereas CL enforces stability-plasticity equilibrium across sequential tasks. To mitigate catastrophic forgetting in CL, we implement elastic weight consolidation (EWC), which imposes quadratic constraints on parameter updates proportional to their Fisher information importance (Kirkpatrick et al., 2017). Specifically, a regularization term L_R is incorporated into the loss function to minimize the rate of change in critical parameters:

$$L_R(\theta) = \frac{\lambda}{2} \sum_{i=1}^n F_i^* (\theta_i - \theta_i^*)^2 \quad (13)$$

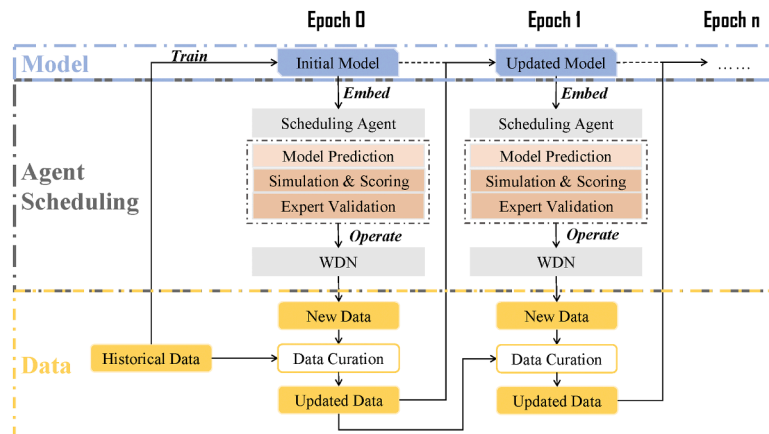


Fig. 3. Workflow of the data-model coevolution cycle.

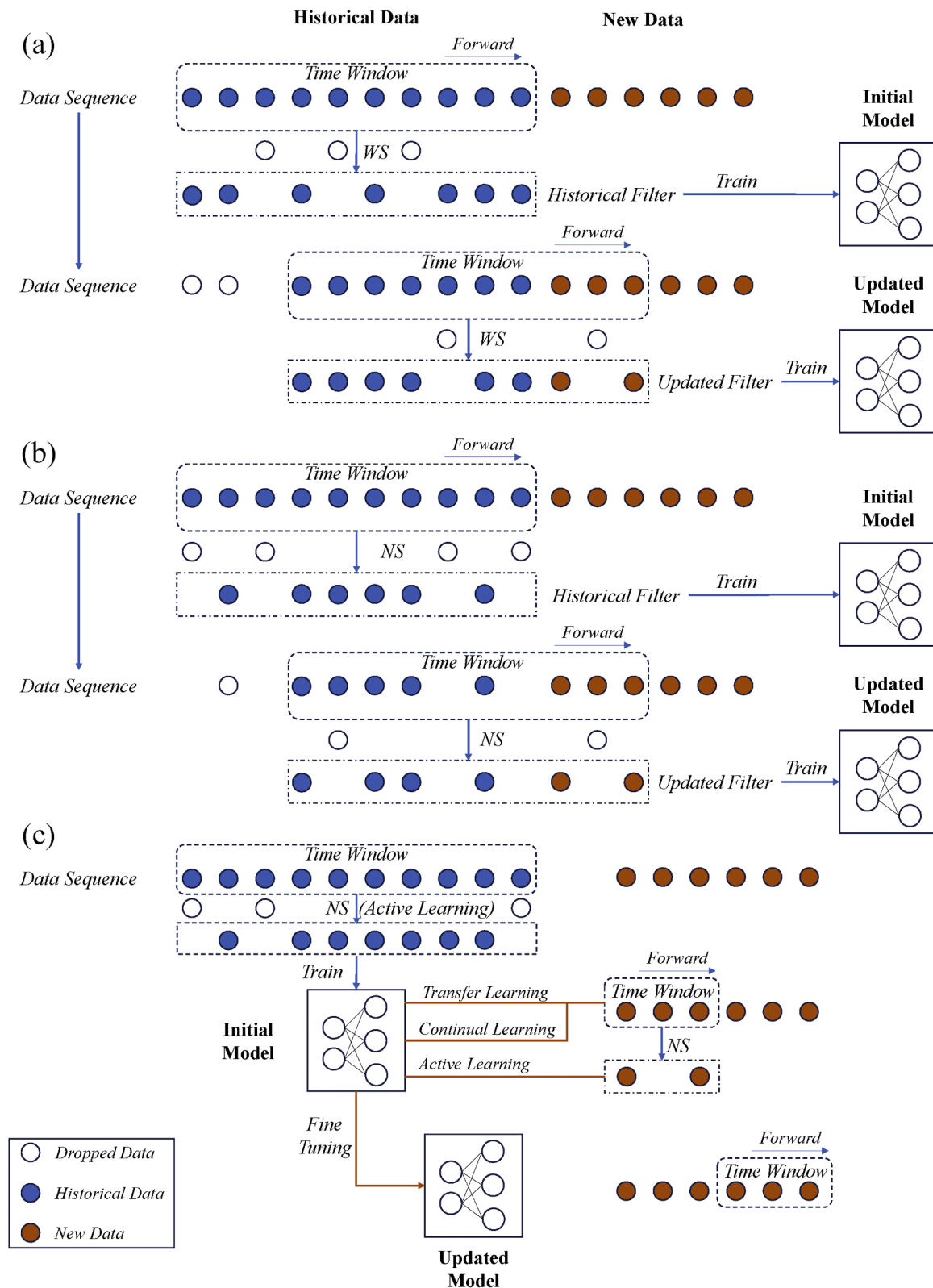


Fig. 4. The process of data-model coevolution: (a) DM-WS; (b) ES-NS; (c) TL, CL and AL.

$$F_i^* = \frac{1}{|D^*|} \sum_{d^* \in D^*} \frac{\partial^2 L^*(d^*, \theta_i^*)}{\partial \theta_i^{*2}} \quad (14)$$

where n denotes the total number of trainable parameters in the neural network; θ_i^* represents the i th parameter trained to convergence on the

old task, and θ_i corresponds to the i th parameter after fine-tuning for the new task; the importance of the i th parameter, F_i^* , is quantified by the average second-order derivative of the parameter over all samples; d^* denotes a sample from the old dataset D^* .

As shown in Fig. 4(c), we implemented a rolling-update mechanism for scheduling model optimization based on parameter fine-tuning. The

application of TL, Method (3), and CL (with EWC), Method (4), were implemented comparatively to evaluate their differential impacts.

To further enhance model updating, we incorporate the data curation system into fine-tuning and devise an active learning (AL)-based strategy (Lewis, 1994). AL initially trains a model with a limited number of samples, then employs an automated query strategy (Kumar et al., 2020) to select the most informative samples from a large pool based on the model's predictions on unseen data, thereby improving training efficiency. In this study, we first train an initial model using historical data (either the full dataset or a curated subset of high-quality samples). Subsequently, we apply the NS method (Section 2.3.2 (2)) to select high-quality samples from incoming data (i.e., intelligent scheduling decisions) and fine-tune the model in conjunction with the EWC method. This integrated approach, designated NS-EWC, combines data curation (NS) with continual learning (EWC) and aims to accelerate performance gains using only a small amount of high-quality data.

3. Results and discussion

This study uses operational data from the eastern Qingpu District WDN in Shanghai, China, which serves approximately 699,000 residents. The system is supplied by a water treatment plant with a capacity of 200,000 m³/day. Fig. 5(a) presents a schematic topology of the regional network, which includes over 38,000 junctions and 31,000 pipes arranged in a looped-mesh configuration. Continuous hydraulic monitoring is provided by an extensive SCADA network comprising >20 flowmeters and a similar number of pressure sensors. The dataset has a 5-minute temporal resolution and covers data from April 2020 onward.

3.1. Dataset splitting

As shown in Fig. 5(b), the testing sets are categorized as follows:

Category 1: Recent data to test model performance under normal operating conditions. Unlike prediction tasks, decision-making tasks for generating scheduling strategies lack a unique ground truth, as manual scheduling strategies are not always optimal. Therefore, it is necessary to select benchmark periods with demonstrably good scheduling performance for comparative evaluation. In addition, both high- and low-demand scenarios should be covered to validate robustness. Accordingly, we selected Testing Set 1 (2023.8.18–2023.8.26) and Testing Set 2 (2023.2.22–2023.3.2). Detailed data analysis and selection rationale are provided in the Supplementary Material.

Category 2: Historical data to test the models' retention capacity for historical scenarios after updates. A historical period outside the training set was randomly selected (Testing Set 3, 2021.5.1–2021.5.9).

The original training sets are categorized as follows:

Category 1: Historical data for the training of initial models (Training Set 1, 2020.4.18–2021.4.30).

Category 2: Newly collected data for the rolling update of models (Training Set 2, 2022.6.20–2022.8.28), comprising five sequential 14-day batches collected during deployment of the scheduling agent, under both model prediction and technicians' empirical judgment. These data are used sequentially to update the initial models.

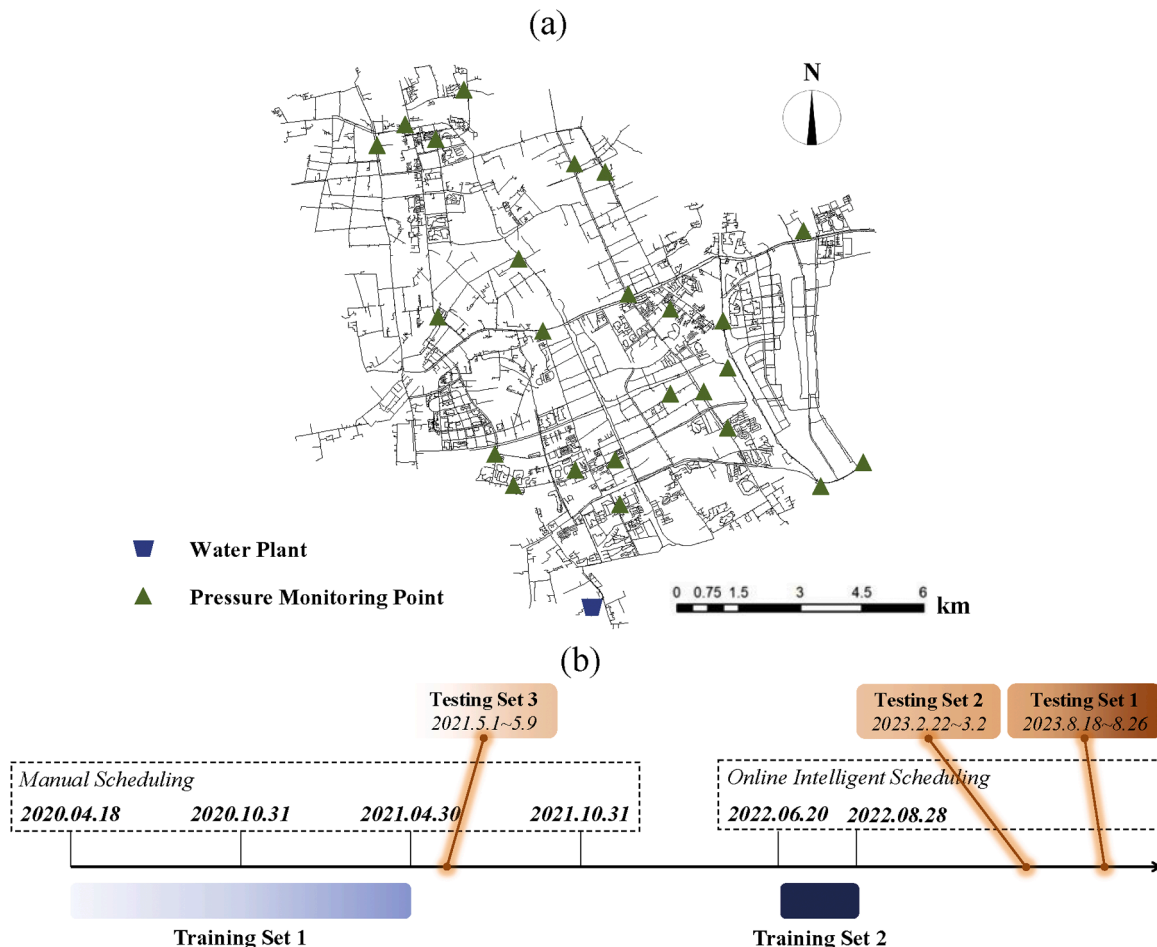


Fig. 5. Data source and split: (a) Topological structure of the WDN; (b) Time ranges of testing sets and training sets.

The detailed statistical features of the training and testing sets, along with the rationale for their split, are presented in the Supplementary Material.

3.2. Experimental design of model evolution

At the stage of rolling upgrading models through the data-model coevolution mechanism, the design of grouping experiments and repeated experiments is shown in Table 1. The ES-NS methodology utilizes a curated dataset of 10,000 samples, a volume equivalent to the filtered dataset obtained by DM-WS. For DM-WS and ES-NS, five repeated training runs are conducted at each stage. For TL, CL, and AL, the five models obtained from the initial stage (via repeated experiments) are each continuously fine-tuned.

Fig. 6 presents the time consumption of each operation in the data curation system. The initial model establishment phase is performed offline, and the time consumption of all operations remains within an acceptable range. In the Model evolution phase, ES-NS only needs to add the incoming new data to the elite sample pool and perform NS on these data, rather than on the entire dataset. Consequently, the time consumption is significantly reduced, facilitating online updates in this phase.

3.3. Test error analysis

The evolution of model performance during the updating process is reflected by two test error metrics, MAE and RMSE, with results presented in Fig. 7. Fig. 7(a) illustrates the distribution of MAE across repeated experiments at different update stages for various updating methods. In repeated experiments, the randomness for DM-WS and ES-NS primarily stems from the initial random weights during each retraining. For TL and CL, which only fine-tune the parameters of the previous model during the update stage, the randomness mainly originates from the different weights of the initial models obtained from repeated experiments. It can be observed that the initial models exhibit large MAE variance, indicating pronounced training randomness. Models trained on the same batch of data yield inconsistent performance across different training runs. This suggests that the historical data suffer from poor consistency and fail to meet the data requirements of DCAI.

After rolling updates, both DM-WS and ES-NS methods reduce training randomness and enhance model training stability, indicating improved sample consistency. We examine the rolling optimization of the dataset achieved by these two mechanisms. DM-WS attains a 3.99% improvement in PSO fitness (from 2.056 to 2.138). Meanwhile, for the ES-NS population, the average data score (overall mean of s_1 and s_2 , as defined in Section 2.3.1) rises from 0.9841 to 0.9857. These findings demonstrate that DM-WS and ES-NS enhance data quality through the iterative integration of new and old data, thereby driving the evolution of DCAI's datasets.

For the fine-tuning based TL and CL methods, the MAE variance among the five initially disparate models gradually diminishes, and their performance converges. This indicates that these two methods provide targeted and controllable model optimization, and the rolling update process is robust to differences in the initial models.

Fig. 7(b) and (c) respectively display the average MAE and RMSE of different models at each update stage, reflecting the overall performance

trends. MAE and RMSE exhibit similar patterns. Given the operational tolerance threshold of ± 5 kPa for pressure adjustments (relative to a nominal control variation of 10 kPa), models achieving MAE < 5 kPa are considered operationally viable. Under high-demand conditions (Testing Set 1), TL and CL achieve rapid operational readiness, with MAE reaching 4.64 kPa and 4.92 kPa respectively by Update-1, outperforming ES-NS, which attains compliance at Update-4 (4.42 kPa). DM-WS lags significantly, achieving threshold compliance only by Update-5 (4.69 kPa). After five update cycles, the final MAE optimization rates for TL, CL, ES-NS, and DM-WS are 71.03%, 64.79%, 61.12%, and 45.14%, respectively.

Conversely, low-demand scenarios (Testing Set 2) reveal inverse efficacy patterns: ES-NS and DM-WS meet error thresholds at Update-1 (3.97 and 4.37 kPa), while TL and CL fail entirely (5.05 and 5.09 kPa at Update-5, respectively). After five update cycles, the final MAE optimization rates for TL, CL, ES-NS, and DM-WS are 36.29%, 35.75%, 49.04%, and 47.76%, respectively. This performance dichotomy correlates with the compositional emphasis of Training Set 2 on high-demand patterns. This characteristic preferentially enhances the adaptation of fine-tuning methods to new scenarios, while retraining approaches buffer data shifts through the retention of historical knowledge.

Notably, Testing Set 3 evaluations expose critical memory retention trade-offs. Retraining methods (ES-NS/DM-WS) effectively preserve historical operational wisdom. Fine-tuning strategies exhibit divergent behaviors: TL progressively erodes historical knowledge (MAE = 5.62 by Update-4), while CL's regularization mechanisms mitigate catastrophic forgetting at the expense of slower new pattern assimilation.

Three operational insights emerge: retraining methods balance historical preservation with incremental adaptation, fine-tuning prioritizes rapid new scenario mastery, and ES-NS uniquely reconciles these competing demands. With 32.17% faster high-demand convergence than DM-WS (1.14 vs. 0.77 kPa/update cycle) and 95.08% better historical retention than CL ($\Delta 0.15$ vs. 3.07 kPa at Update-5), ES-NS proves optimal for iterative model refinement, allowing the agent to adaptively update while retaining historical knowledge.

3.4. Energy consumption and pressure control performance

Test errors reflect the proximity of model predictions to the benchmark test set, serving as indicators of theoretical model performance. However, to ensure that models meet practical application requirements, it is essential to evaluate the impact of the predicted actions on the WDN through hydraulic simulation. The validation involves two categories of metrics: (1) water supply safety indicators, including nodal pressure satisfaction rate (%) and pressure fluctuation rate (%); and (2) energy consumption indicator, characterized by unit energy consumption (kWh/km^3). Test results on the high-demand testing set (Testing Set 1) and low-demand testing set (Testing Set 2) are presented in Fig. 8(a) and (b), respectively. Repeated experiments were conducted for model training within each update cycle. Reference lines (dashed lines in the figures) represent the corresponding metric values of the test sets.

It can be observed that, compared with TL and CL, the historical models of DM-WS and ES-NS generate scheduling schemes with relatively higher energy consumption but achieve better performance in the two water supply safety indicators: pressure fluctuation rate and pressure satisfaction rate. This suggests that filtering and optimizing historical datasets facilitates early deployment of the models.

On the high-demand test set, as update cycles progress, DM-WS and ES-NS substantially reduce energy consumption in intelligent scheduling (DM-WS: 5.22% reduction; ES-NS: 4.05% reduction, both below the energy consumption level of the test set) and nodal pressure fluctuation rate (DM-WS: 4.77% reduction; ES-NS: 4.07% reduction). However, DM-WS compromises pressure satisfaction rate, achieving an average of 96.60% by the fifth update cycle, whereas ES-NS maintains a high pressure satisfaction rate of 99.45% at the fifth cycle. TL and CL show no significant optimization in energy consumption; their improvements

Table 1
Experimental design of model evolution.

Training Set #	Method Category	Method	Experimental Times
Training Set 1 & 2	Retraining	DM-WS	5
		ES-NS	5
Training Set 2	Fine-tuning	TL	5
		CL (EWC)	5
		AL(NS-EWC)	5

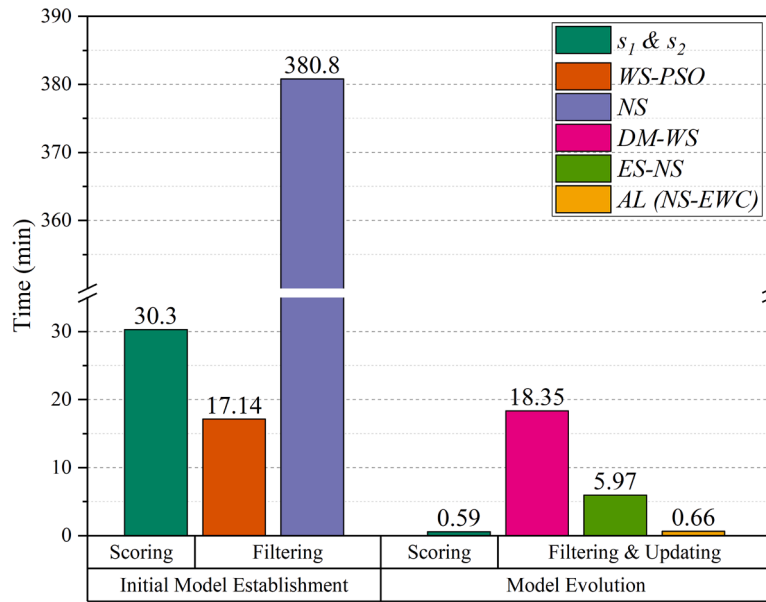


Fig. 6. Time consumption of the data curation process.

primarily focus on pressure fluctuation rate (TL: 4.73% reduction; CL: 2.28% reduction) and pressure satisfaction rate (TL: 3.62% increase; CL: 3.54% increase).

On the low-demand test set, the overall trends of the methods are similar to those on the high-demand test set, still demonstrating optimization effects, albeit with reduced magnitudes. A possible explanation is that the newly added data originate from periods of high water demand within the year, with limited coverage of low-demand samples. TL and CL, which update models using only newly added data, experience greater disruptive impacts; the energy consumption of schemes generated by the updated models increases slightly (TL: 0.66% increase; CL: 0.14% increase). In contrast, DM-WS and ES-NS retain historical data encompassing multiple operating conditions, enabling them to consistently exert optimization effects. Notably, on the low-demand test set compared to the high-demand test set, DM-WS and ES-NS achieve reductions in energy consumption (DM-WS: 2.86% reduction; ES-NS: 2.28% reduction) and pressure fluctuation rate (DM-WS: 4.75% reduction; ES-NS: 4.33% reduction) while better maintaining pressure satisfaction rate, which remains above 99% throughout the update process. This indicates that high-demand period data containing only a few low-demand samples can still effectively optimize model performance under low-demand conditions. Moreover, due to the conservative scheduling strategies during high-demand periods, the models maintain stable water supply safety. This demonstrates the robustness of the updating methods across different update time points.

Overall, the ES-NS method simultaneously achieves the two critical objectives of reducing energy consumption and improving water supply safety, making it the most suitable updating method for practical application. This suggests that maintaining a Pareto-optimal set that comprehensively considers all indicators during the update process facilitates holistic model optimization. Although DM-WS exhibits theoretical performance (characterized by test errors) comparable to ES-NS, it may underperform in certain indicators during practical deployment, possibly related to the tendency of heuristic algorithms to fall into local optima, with the fitness function design exerting considerable influence. TL and CL, which rely solely on unoptimized newly added data, show slightly inferior optimization performance. Nevertheless, they still achieve improvements in water supply safety while maintaining similar energy consumption levels, substantially enhancing the practical usability of the agents.

3.5. Model interpretability analysis

This study employs SHAP values to analyze model interpretability, investigating whether the continuous optimization of models via updating methods can be explained by changes in internal mapping relationships (input contributions) and whether this optimization is directionally controllable, thereby enhancing the credibility of the optimization approach. The ES-NS method, which achieved the best optimization performance, is selected as a representative for analyzing the importance of model inputs (characterized by average SHAP values) at different update stages. The top five most important inputs for each model are extracted, with results presented in Fig. 9.

It can be observed that the recent monitoring value of the water treatment plant outlet flow, specifically WP Pressure (GRU) representing information extracted by the GRU layer, gradually becomes the most important input exerting the greatest influence on the output. This variable represents the current real-time water demand, serving as the most critical monitoring value reflecting the real-time water supply status of the network. The increased importance of this input indicates that the model can more accurately perceive current operating conditions and better identify trends in network state changes. Furthermore, the importance of historically corresponding monitoring values of water treatment plant outlet flow and pressure (WP Flow (Conv) and WP Pressure (Conv), which are extracted by the convolutional layer) also progressively increases. These two inputs reflect water supply patterns of the plant during identical time periods across adjacent scheduling cycles (daily) and across weekly scheduling cycles, encompassing historical scheduling schemes and operating experience. The increased importance of these inputs suggests that the model can more effectively capture key patterns from historical data and learn the essential experience embedded within.

Consequently, ES-NS optimization not only improves test errors and simulation metrics but also aligns the model's internal mappings with objective laws and enhances the use of deep learning's data mining capabilities. This dynamic interpretability analysis strengthens both the directionality and credibility of the agent's self-evolution.

3.6. Performance of active learning

Guided by the active learning strategy, this study uses the NS algorithm to screen high-quality samples, creating separate elite sample

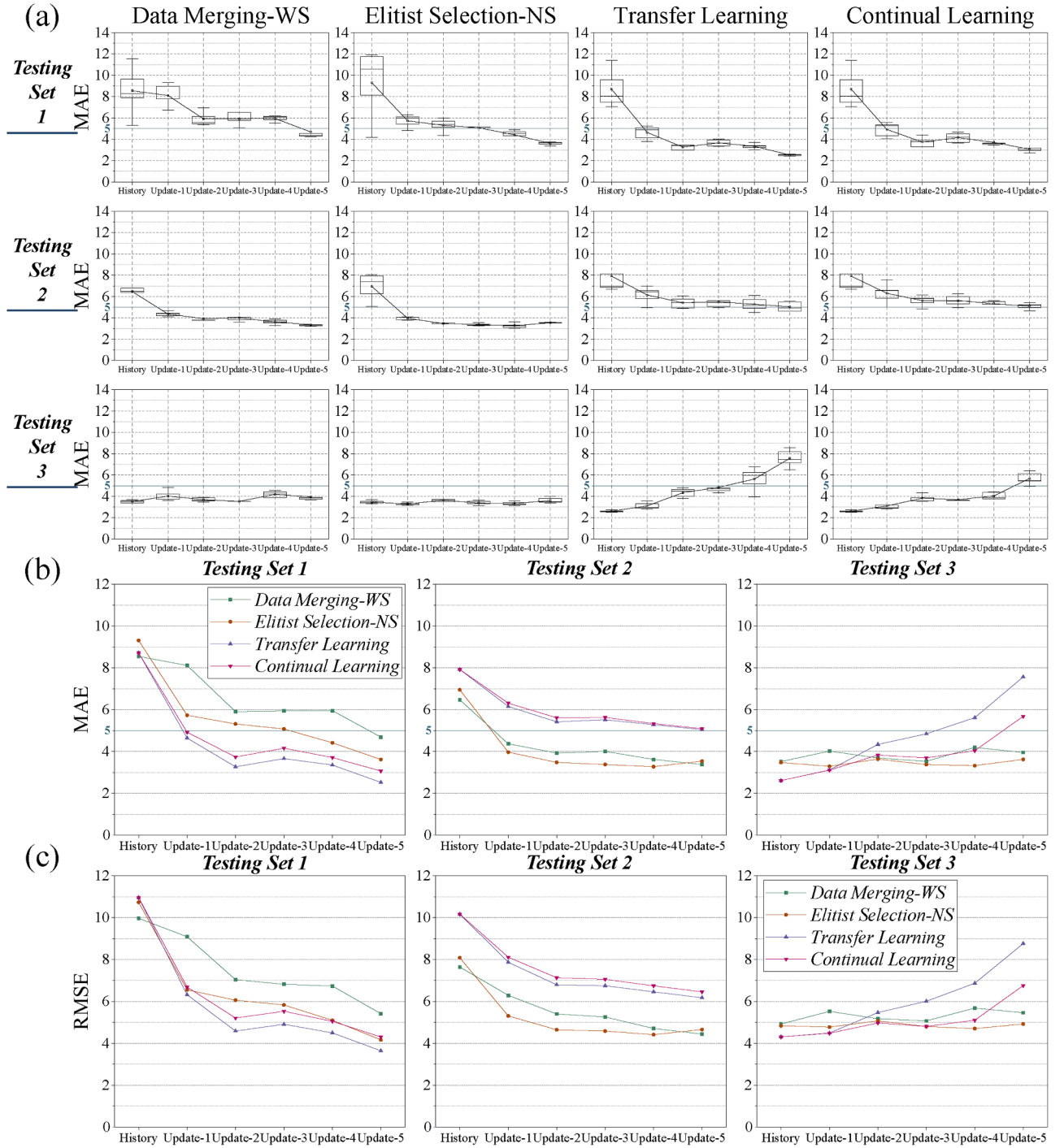


Fig. 7. Test error performance of models during rolling update: (a) MAE distributions; (b) MAE mean values; (c) RMSE mean values.

pools for historical and incoming data at each update stage. Based on the quantity ratios between historical and newly added data pools, several groups were designed for repeated experiments: all historical data with 2000 newly added samples (all: 2000), all historical data with 3000 newly added samples (all: 3000), 50,000 historical samples with 2000 newly added samples (50,000: 2000), 50,000 historical samples with 3000 newly added samples (50,000: 3000), 20,000 historical samples with 2000 newly added samples (20,000: 2000), and 20,000 historical samples with 3000 newly added samples (20,000: 3000). The MAE test results (averages of repeated experiments) of this series of AL tests at each update stage are compared with TL and CL methods using all data (all: all), as presented in Fig. 10. The left panel plots MAE across stages

on a shared axis to illustrate overall trends, while the right panel employs independent axes for each stage to facilitate numerical comparison among different methods.

It can be observed that when the AL initial model is trained using all historical data (all: 3000), the optimization efficiency of AL is inferior to that of CL: after five update cycles, the MAE reduction of AL (all: 3000) is 3.77% lower than that of CL on Test Set 1 and 7.78% lower on Test Set 2. The two methods exhibit similar performance on Test Set 3. This indicates that when fine-tuning models, the quantity of newly added data substantially influences optimization effectiveness, particularly when the initial model is well-trained. In such cases, the quantity of new data deserves greater attention than its quality.

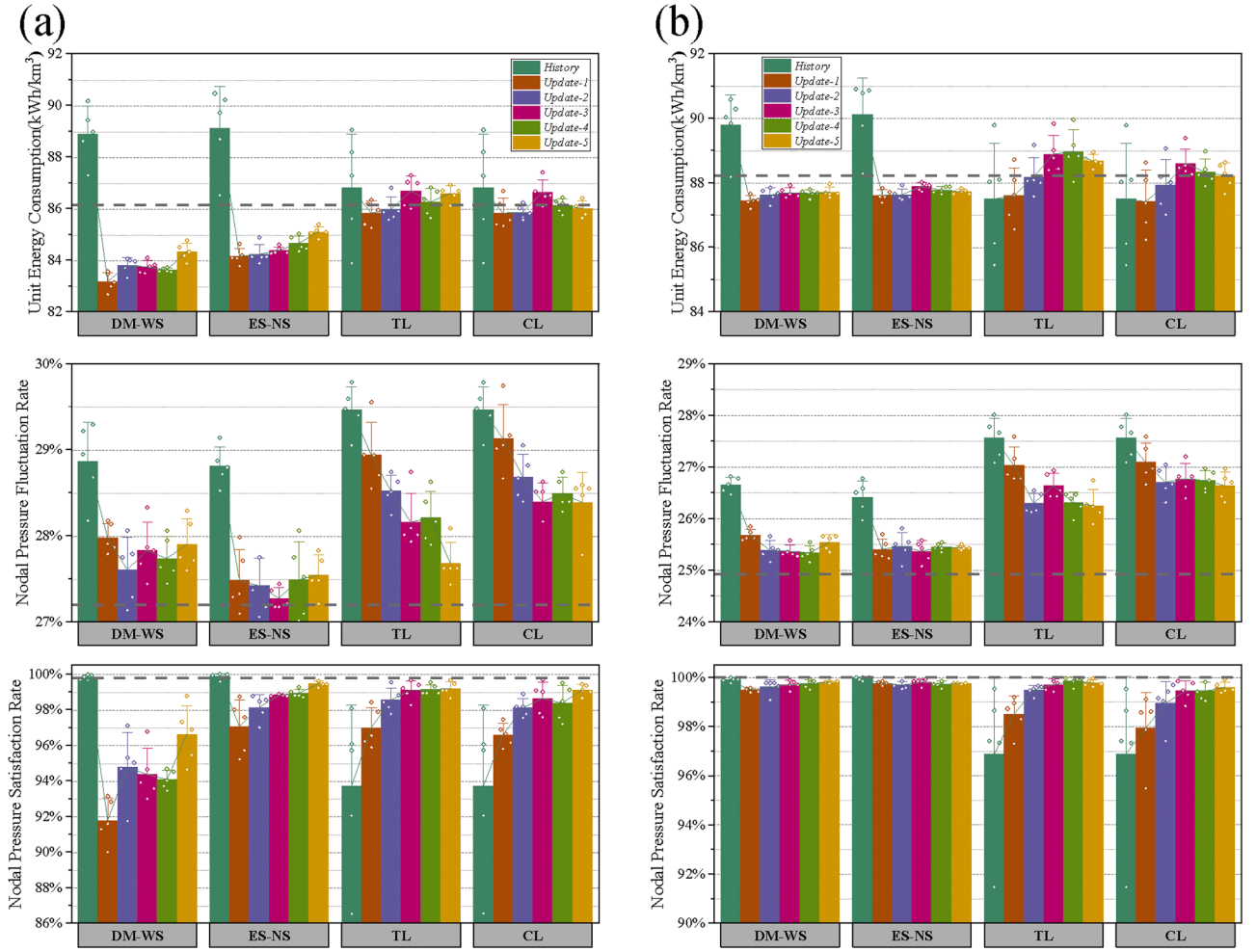


Fig. 8. Evolution of model scheduling performance during rolling updates: (a) Testing set 1; (b) Testing set 2.

When the AL initial model is trained using elite data filtered by the NS method (50,000: 3000 and 20,000: 3000), AL demonstrates improved optimization effects on Test Sets 1 and 2 compared to CL. Notably, AL with a more refined initial training dataset (20,000: 3000) achieves superior optimization results. After five updates, on Test Set 1, AL (20,000: 3000) reduces MAE to 2.88 kPa (optimization rate: 57.32%), which is 6.00% lower than CL and approaches the MAE of 2.52 kPa achieved by TL. On Test Set 2, the advantage of AL (20,000: 3000) is even more pronounced, reducing MAE to 4.71 kPa (optimization rate: 25.04%), which is 7.45% lower than CL and 6.66% lower than TL. Remarkably, AL (20,000: 3000) is the only method among those compared that successfully optimizes the model to meet the operational standard (MAE < 5 kPa) on Test Set 2, achieving this as early as the first update (MAE = 4.88 kPa). However, on Test Set 3, the MAE of AL (20,000: 3000) increases rapidly throughout the update process, reaching 10.72 kPa after five updates, exceeding the 7.57 kPa of TL. This suggests significantly increased forgetting of historical data, rendering the method no longer applicable to historical operating conditions. This indicates that if the initial model is undertrained (with limited training data), it may compromise the memory retention capacity of EWC-based fine-tuned models regarding historical scenarios.

Consequently, the performance of AL depends on the ratio between initial training data and newly added data. Tuning this ratio allows AL to emulate either optimized TL or CL. A well-trained initial model (trained on abundant data) makes the quantity of new data the key factor in fine-tuning; moderately reducing initial data enables AL to accelerate optimization beyond CL. With less initial data, AL achieves better

adaptation to new scenarios and outperforms TL in learning minority operational patterns from incoming data, albeit at the risk of increased forgetting of historical knowledge. The flexibility of AL facilitates interactive collaboration between the agent and technicians, enabling the achievement of different evolution goals.

4. Conclusions

DCAI enhances the optimality of predictive outcomes by learning human expertise from high-quality datasets, thereby enabling real-time decision-making for scheduling actions in WDNs. An intelligent scheduling agent deployed with DCAI can achieve efficient and reasonable pump scheduling through real-time human-computer interaction between technicians and AI models. To empower the scheduling agent with autonomous online updates after deployment, this study proposes a DCAI continuous learning method based on data-model coevolution. This approach transforms the model from a static predictor into a self-evolving autonomous agent capable of adapting to new operating conditions and continuously optimizing scheduling performance.

The data-model coevolution mechanism drives iterative performance refinement, with elitist selection-driven retraining (ES-NS) significantly reducing pressure prediction errors in high-demand scenarios (Testing Set 1): mean MAE decreased by 61.12% while variance decreased by 99.81%, demonstrating exceptional model enhancement. Furthermore, ES-NS demonstrates robust historical scenario retention, exhibiting minimal performance degradation (MAE increase: 4.35%) when validated against legacy operational datasets. Comparatively, TL

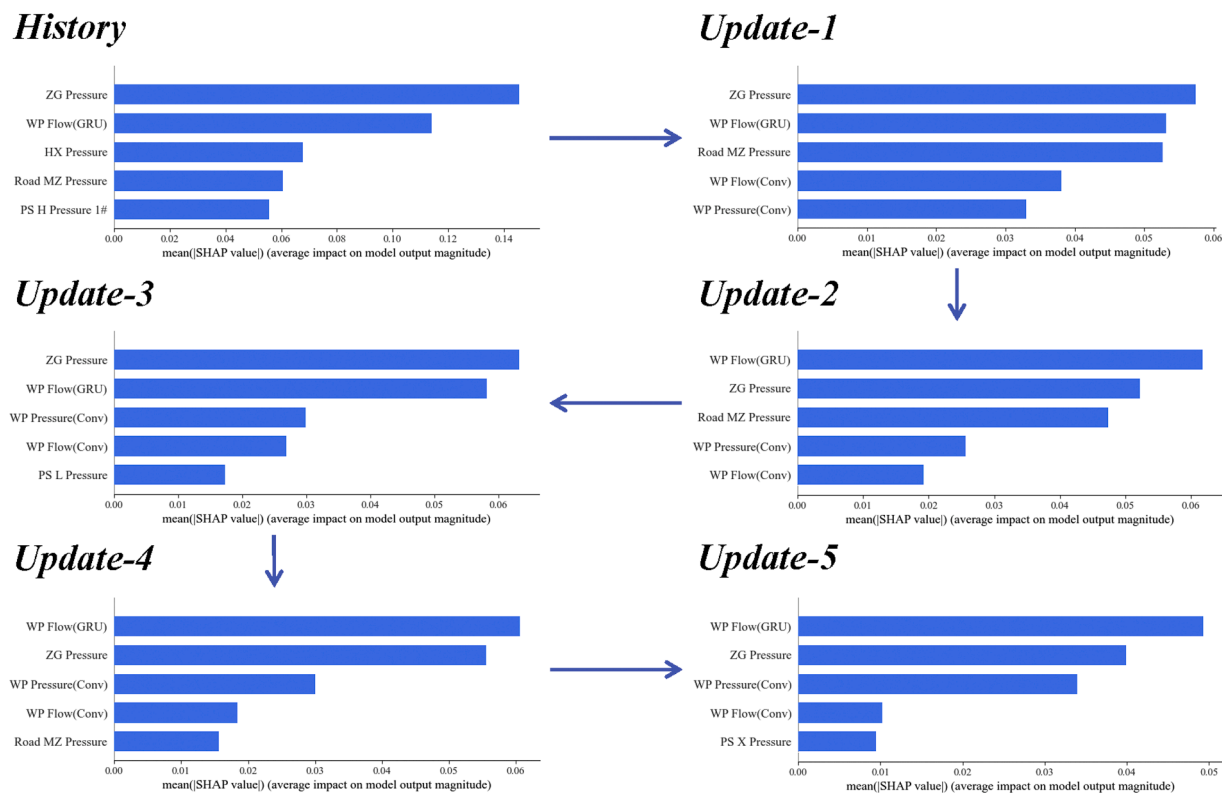


Fig. 9. Changes in input importance ranking of models obtained by the ES-NS method at different update stages (Top 5 inputs).

achieves rapid new scenario adaptation (MAE from 8.71 to 2.52 kPa) but exhibits substantial forgetting of historical scenarios (MAE from 2.61 to 7.57 kPa), while CL preserves memory at the cost of 8.79% slower new pattern assimilation. When validating the energy consumption control and pressure control performance of agent-generated scheduling schemes via hydraulic simulation, the ES-NS method simultaneously achieves the objectives of reducing energy consumption and ensuring water supply safety, demonstrating high adaptability for practical engineering applications. TL and CL offer limited energy consumption optimization but enhance scheme safety, still meeting fundamental requirements for practical deployment. Furthermore, observing the trends in model interpretability analysis during ES-NS optimization reveals that the internal mapping relationships of the model become progressively more aligned with objective laws throughout the update process, validating the directional controllability of the optimization method. These trade-offs position ES-NS as the preferred strategy for agents requiring historical continuity. For scenarios with data security constraints or unavailable historical data, model fine-tuning provides an optimal approach: employ TL when historical patterns become obsolete, and CL for persistent operational contexts.

Based on CL (EWC), this study develops an AL-based model updating method (NS-EWC), which employs NS as the query strategy to utilize newly added data more efficiently and shorten the time window required for updates. Experimental results show that the effectiveness of AL is substantially influenced by the ratio between the amount of initial training data and the newly added data. When the initial model is trained on a relatively small dataset, AL adapts more rapidly to new operating conditions during the update phase. It particularly excels at capturing and learning minority patterns within the incoming data, achieving overall performance superior to that of TL, albeit with a comparable level of historical forgetting. When the volume of initial training data is moderately increased, AL exhibits enhanced memory retention, attaining overall performance that surpasses that of CL. Therefore, the AL updating strategy can be flexibly adjusted in practical engineering applications according to specific requirements.

Through the cyclical data-model coevolution enabled by the data curation system and model updating methods, DCAI is transformed into an autonomous scheduling agent that perceives network states, executes decisions, and continuously evolves through environmental interaction. This contributes to more resilient, efficient, and sustainable urban water management. However, the feedback that the current agent can process is primarily derived from WDN responses. Embedding LLMs into the agent to handle textual feedback from technicians represents a critical direction for future research. Other research directions extendable from this work include: multi-agent systems (MAS) for network-wide coordination; hybrid AL query strategies that couple uncertainty quantification with performance-based filtering to improve sample efficiency for rare operational patterns; hierarchical deep reinforcement learning for exploring superior decisions beyond historical expertise while building upon human experience; incorporation of time-of-use electricity tariffs into the optimization objectives for cost-aware scheduling; and systematic investigation of the diminishing marginal returns in model updates along with long-term field validation to confirm sustained stability.

CRediT authorship contribution statement

Minghai Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhengheng Pu:** Software, Project administration, Investigation. **Hexiang Yan:** Project administration, Investigation, Funding acquisition. **Tao Tao:** Project administration, Investigation, Funding acquisition. **Kunlun Xin:** Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

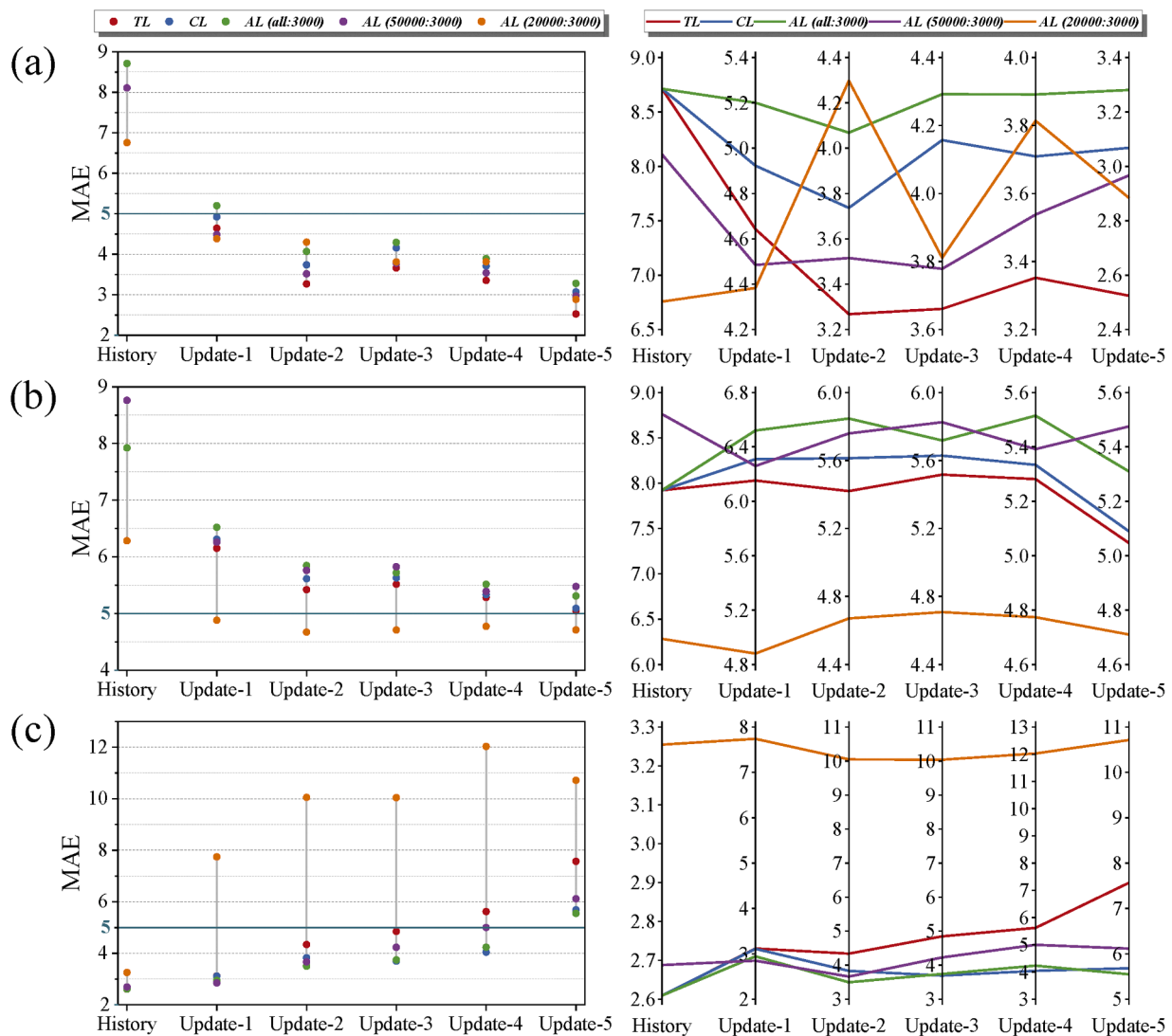


Fig. 10. Evolution and comparison of MAE for AL, TL, and CL models during rolling updates: (a) Testing set 1; (b) Testing set 2; (c) Testing set 3.

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2026.126284](https://doi.org/10.1016/j.watres.2026.126284).

Data availability

The data that has been used is confidential.

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